

HIGH-VOLTAGE GENERATOR For Microcontroller Projects

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To drive small devices like neon lamps, you need a high-voltage DC power supply. This circuit is a simple DC-to-DC boost (step-up) converter for general-purpose applications. It uses readily available parts that can be bought from any electronic components store. The author's prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the high-voltage generator is shown in Fig. 2. It is built around NE555 timer (IC1), STP55NF0 MOSFET (IRF1), diode UF4001 and a few other components. Values of resistor R4 and Zener diode ZD1 are based on the voltage and current required for

your application.

NE555 timer is configured as an astable multivibrator oscillator, which oscillates at a frequency (around 68kHz) determined by the values of resistors R1, R2 and capacitor C1. Its output directly drives a MOSFET (IRF1) that switches the current through a 10 μ H/200mA inductor (L1).

Fast recovery diode D1 and high-voltage capacitor C4 together provide an efficient output rectifier and filter circuit, while R3 and C3 forms a snubber network for IRF1.

Place the components in appropriate places on the PCB. Connect a 5V

DC source to input CON1. Check the output voltage available at output connector CON2 using a multimeter—it could be anywhere between 30V DC and 90V DC. You may trim it for real-world applications by adding a simple Zener diode-based shunt regulator circuitry, as indicated within dotted lines in the circuit diagram. Note that, final output voltage varies with a connected load, but it should not swing by more than a few volts for a load up to 10mA.

Example. Assume that raw output voltage across CON2 is 50V (V_{in}), and what you are looking for is 1mA

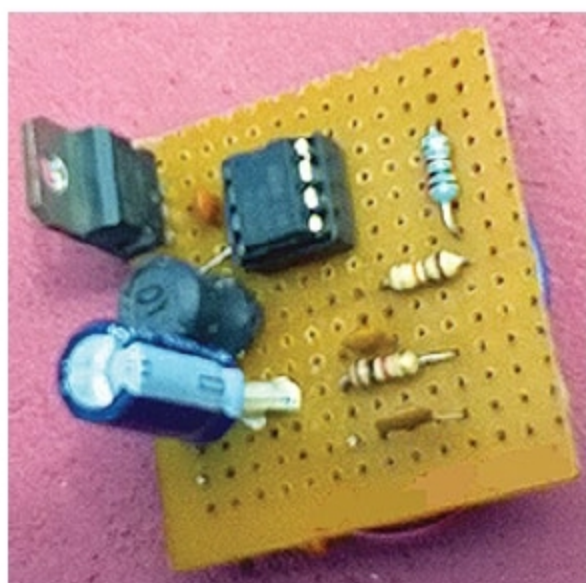


Fig. 1: Author's prototype

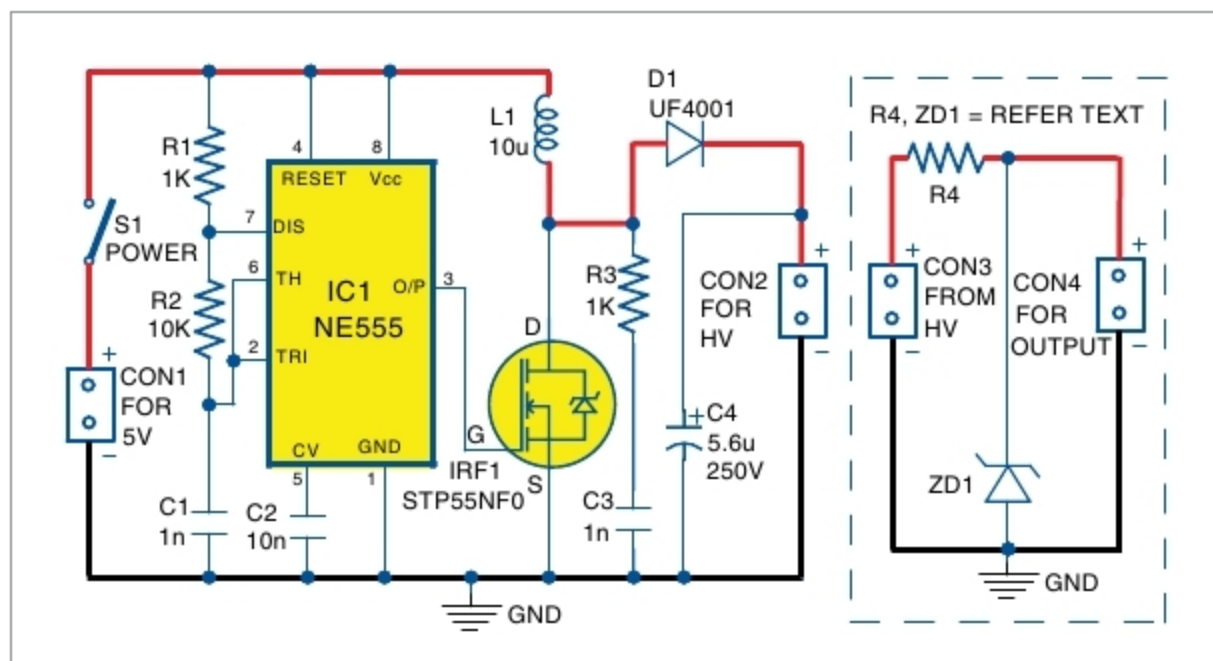


Fig. 2: Circuit diagram of high-voltage generator for microcontroller projects

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